

2. Dik: Perusahaan mengadopsi enkripsi sederhana menggunakan perkalian Matriks

Setiap blok 3×3 dari citra logo dienkripsi dengan $E = A \cdot P$

Dimana A Matriks Kunci dan P blok pixel. Dekripsi dengan $P = A^{-1} \cdot E$

$$\text{Matriks } A = \begin{bmatrix} a+1 & b & 2 \\ 1 & a+2 & b \\ b & 1 & a+3 \end{bmatrix}$$

a.) Tulis Matriks A dan Hitung Det A

= Substitusi:

$$A = \begin{bmatrix} 2 & 6 & 2 \\ 1 & 3 & 6 \\ 6 & 1 & 4 \end{bmatrix}$$

$$\text{Det } A = \begin{vmatrix} 2 & 6 & 2 \\ 1 & 3 & 6 \\ 6 & 1 & 4 \end{vmatrix} = \begin{vmatrix} 2 & 6 \\ 1 & 3 \end{vmatrix} - 2 \begin{vmatrix} 1 & 6 \\ 6 & 4 \end{vmatrix} + 2 \begin{vmatrix} 1 & 3 \\ 6 & 1 \end{vmatrix}$$

$$\begin{aligned} \text{Det } A &= 2(3 \cdot 4) - (6 \cdot 1) - 6(1 \cdot 4) - (6 \cdot 6) + 2(6 \cdot 3) \\ &= 2 \cdot 6 - 6(-32) + 2(-17) \\ &= 12 + 192 - 34 \end{aligned}$$

$$= 170$$

β. Hitung semua Minor

$$M_{11} = \begin{vmatrix} 3 & 6 \\ 1 & 4 \end{vmatrix} = (3 \cdot 4) - (6 \cdot 1) = 12 - 6 = 6$$

$$M_{12} = \begin{vmatrix} 1 & 6 \\ 6 & 4 \end{vmatrix} = (1 \cdot 4) - (6 \cdot 6) = 4 - 36 = -32$$

$$M_{13} = \begin{vmatrix} 1 & 3 \\ 6 & 1 \end{vmatrix} = (1 \cdot 1) - (3 \cdot 6) = 1 - 18 = -17$$

$$M_{21} = \begin{vmatrix} 6 & 2 \\ 1 & 4 \end{vmatrix} = (6 \cdot 4) - (2 \cdot 1) = 24 - 2 = 22$$

$$M_{22} = \begin{vmatrix} 2 & 2 \\ 6 & 4 \end{vmatrix} = (2 \cdot 4) - (6 \cdot 2) = 8 - 12 = -4$$

$$M_{23} = \begin{vmatrix} 2 & 6 \\ 6 & 1 \end{vmatrix} = (2 \cdot 1) - (6 \cdot 6) = 2 - 36 = -34$$

$$M_{31} = \begin{vmatrix} 6 & 2 \\ 3 & 6 \end{vmatrix} = (6 \cdot 6) - (2 \cdot 3) = 36 - 6 = 30$$

$$M_{32} = \begin{vmatrix} 2 & 2 \\ 1 & 6 \end{vmatrix} = (2 \cdot 6) - (2 \cdot 1) = 12 - 2 = 10$$

$$M_{33} = \begin{vmatrix} 2 & 6 \\ 1 & 3 \end{vmatrix} = (2 \cdot 3) - (6 \cdot 1) = 6 - 6 = 0$$

$$A^{-1} = \frac{1}{\text{Det } A} \begin{bmatrix} 6 & -32 & -17 \\ 22 & -4 & -34 \\ 30 & 10 & 0 \end{bmatrix}$$

(KKY)

C. Matriks kofaktor dan invers

$$\text{Cofactor} = \begin{bmatrix} 6 & 32 & -17 \\ -22 & -4 & 34 \\ 30 & -10 & 0 \end{bmatrix}, \quad \text{Adj}(A) = \begin{bmatrix} 6 & -22 & 30 \\ 32 & -4 & -10 \\ -17 & 34 & 0 \end{bmatrix}$$

$$\text{invers} = \frac{1}{170} \begin{bmatrix} 6 & -22 & 30 \\ 32 & -4 & -10 \\ -17 & 34 & 0 \end{bmatrix}$$

E. Matriks P

$$P = \begin{bmatrix} 120 & (50+N) & 200 \\ 80 & a+2 & (10+N) \\ (30+N) & 90 & 170 \end{bmatrix} = \begin{bmatrix} 120 & 66 & 200 \\ 80 & 3 & 26 \\ 46 & 90 & 170 \end{bmatrix}$$

$E = A \cdot P$ (Enkripsi)

$$= \begin{bmatrix} 2 & 6 & 2 \\ 1 & 3 & 6 \\ 6 & 1 & 4 \end{bmatrix} \cdot \begin{bmatrix} 120 & 66 & 200 \\ 80 & 3 & 26 \\ 46 & 90 & 170 \end{bmatrix}$$

$$1.1 = 2 \cdot (120) + 6(80) + 2(46) = 240 + 480 + 92 = 812$$

$$1.2 = 2 \cdot (66) + 6(3) + 2(90) = 132 + 18 + 180 = 330$$

$$1.3 = 2 \cdot (200) + 6(26) + 2(170) = 400 + 156 + 340 = 896$$

$$2.1 = 1(120) + 3(80) + 6(46) = 120 + 240 + 276 = 636$$

$$2.2 = 1(66) + 3(3) + 6(90) = 66 + 9 + 540 = 615$$

$$2.3 = 1(200) + 3(26) + 6(170) = 200 + 78 + 1020 = 1298$$

$$3.1 = 6(120) + 1(80) + 4(46) = 720 + 80 + 184 = 984$$

$$3.2 = 6(66) + 1(3) + 4(90) = 396 + 3 + 360 = 759$$

$$3.3 = 6(200) + 1(26) + 4(170) = 1200 + 28 + 680 = 1906$$

$$E = \begin{bmatrix} 812 & 330 & 896 \\ 636 & 615 & 1298 \\ 984 & 759 & 1906 \end{bmatrix}$$

$P' = A^{-1} \cdot E$ (Deskripsi)

$$\frac{1}{170} \begin{bmatrix} 6 & -22 & 30 \\ 32 & -4 & -10 \\ -17 & 34 & 0 \end{bmatrix} \cdot \begin{bmatrix} 812 & 330 & 896 \\ 636 & 615 & 1298 \\ 984 & 759 & 1906 \end{bmatrix} = \begin{bmatrix} 120 & 66 & 200 \\ 80 & 3 & 26 \\ 46 & 90 & 170 \end{bmatrix}$$